

WASP-11b

R-1485

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_141029 · created 2026-08-01T13:19:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456959.674 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.051
Transit depth (Rp/Rs)^2	2.94 +/- 1.74 [%]
Orbital Inclination (inc)	86.36 +/- 1.69
Airmass coefficient 1 (a1)	0.86 +/- 0.39
Airmass coefficient 2 (a2)	0.07 +/- 0.049
Scatter in the residuals of the lightcurve fit is	43.94 %
Best Comparison Star	None
Optimal Aperture	1.58
Optimal Annulus	5.0
Transit Duration (day)	0.074 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.051 vs 0.131 (deviation 0.76σ)
Duration fit vs published	0.074 d vs 0.1075 d (deviation 0.93σ)
Mid-time O–C	2.0 min
Depth SNR	3.2
Residual scatter	43.94 %

Light curve

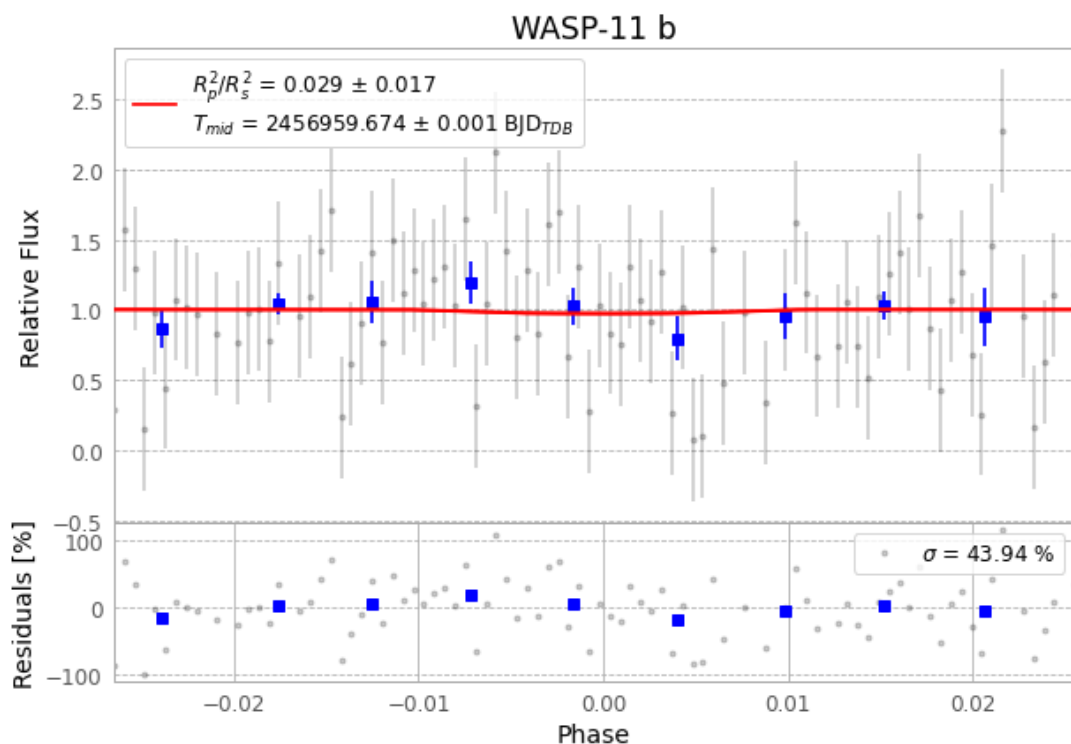


Figure 1 — FinalLightCurve_WASP-11 b_2014-10-28.png (SHA-256 dee6dfe552a86172...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [444, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 44e5ecd3487c5db8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), HATP-10141029013912.FITS → HATP-10141029061812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-141029.log (14839de386a6...), FinalLightCurve_WASP-11 b_2014-10-28.png (dee6dfe552a8...), FinalLightCurve_WASP-11 b_2014-10-28.csv (0cb025e9d7ff...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 13:19Z.